

Fortran 77 Rainfall Simulation Demo from a Random Recent Paper

Automated OpenAlex + Fortran 77 + LaTeX Pipeline

2026-05-18

Selected scholarly paper

Title Permafrost conditions in peatlands govern riverine flushing of dissolved organic carbon, methylmercury, and nutrients

Authors Fares Mandour, Jazmin Greyeyes-Howell, Renae Shewan, Lauren Thompson, Irene Graham, Mike Low, et al.

Source Unknown source

Publication date 2026-05-12

DOI <https://doi.org/10.5194/egusphere-2026-2199>

OpenAlex ID <https://openalex.org/W7160897333>

Landing page <https://doi.org/10.5194/egusphere-2026-2199>

Cited by count 0

Abstract used by the script

Abstract. Permafrost thaw and intensified droughts and floods threaten to alter the mobilization of dissolved organic carbon (DOC), nutrients and methylmercury (MeHg) from boreal peatlands, with cascading impacts on aquatic ecosystem functions and traditional food sources. Here we monitored 27 peatland-dominated (>30 %) catchments in western Canada (150 to 52,000 km²) over a five-year period (2020 – 2024) which captured extreme hydroclimatic conditions. These catchments spanned a climatic and permafrost gradient (mean annual temperature -0.2 to -2.8 °C), which provided a space-for-time framework to assess impacts of continued thaw and warming. Our results demonstrated that catchment climate, and thus permafrost conditions, strongly controlled the hydrological response of solute concentrations. Warmer catchments showed a pronounced flushing response where DOC and MeHg concentrations increased by 50 % and 80 %, respectively, as discharge increased from the 10th to 90th percentile. In contrast, colder catchments maintained a chemostatic response, where concentrations remained stable and low despite hydrological variability. Similar climate-hydrology interactions were found for total nitrogen, total phosphorous, and inorganic phosphorous, but not for inorganic nitrogen. It is likely that permafrost conditions in peatlands affect both the production of solutes and their hydrological connectivity to the stream network. The presence of permafrost in peatlands may act to both e...

Modelling note

This report is an *abstract-driven demonstration*, not a reproduction of the paper's full methods. The script uses the selected title and abstract to tune a small stochastic rainfall-runoff bucket model. It is suitable for software, Fortran 77, and reproducibility demonstrations, not for operational forecasting.

Parameter choices inferred from paper language

- extreme/heavy rainfall language raised storm rainfall mean
- extreme/heavy rainfall language raised heavy-event probability
- extreme/heavy rainfall language raised heavy-event multiplier
- flood language raised direct runoff coefficient
- flood language lowered effective soil storage
- dry/drought language lowered base wet-day probability
- dry/drought language raised evapotranspiration demand
- climate-change/projection language added a rainfall-intensity trend

Parameter	Value
w0	0.2000
pers	0.3500
rmean	14.0000
hprob	0.0800
hfac	4.0000
rcoef	0.3400
etbas	3.8000
cap	75.0000
infil	0.7000
trend	0.1200

Simulation summary

Metric	Value
Days simulated	365
Wet days	149
Total rainfall, mm	1925.50
Total runoff, mm	870.62
Runoff/rainfall ratio	0.452
Maximum daily rainfall, mm	58.83
Maximum daily runoff, mm	46.59
Mean daily rainfall, mm	5.28
Mean daily runoff, mm	2.39

First 10 simulated days

Day	Rain	Runoff	Soil	ET	Wet
1	0.00	0.00	34.11	3.39	0
2	0.00	0.00	30.70	3.41	0
3	0.00	0.00	27.27	3.43	0
4	7.62	2.59	29.14	3.46	1
5	6.17	2.10	29.99	3.48	1
6	0.00	0.00	26.49	3.50	0
7	0.00	0.00	22.96	3.52	0
8	0.00	0.00	19.42	3.54	0
9	0.00	0.00	15.85	3.57	0
10	0.00	0.00	12.27	3.59	0

Generated Fortran 77 source

The generated source file is `rainfall_demo.f`. The daily output is `results.csv`, and the plain-text summary is `summary.txt`.

```

PROGRAM RAIN77
C   RAINFALL/RUNOFF DEMO GENERATED FROM A SCHOLARLY PAPER
C   SELECTED PAPER: Permafrost conditions in peatlands govern riverine flushing of...
C
C   DATA DICTIONARY
C   DAY      = DAY NUMBER, 1 TO 365
C   RAIN     = DAILY RAINFALL DEPTH, MILLIMETRES
C   RUNOF    = DAILY RUNOFF DEPTH, MILLIMETRES
C   SOIL     = SIMPLE SOIL WATER STORE, MILLIMETRES
C   ET       = POTENTIAL EVAPOTRANSPIRATION, MILLIMETRES
C   WET      = 1 FOR WET DAY, 0 FOR DRY DAY
C   WO       = BASE WET-DAY PROBABILITY
C   PERS     = EXTRA WET-DAY PROBABILITY AFTER A WET DAY
C   RMEAN    = MEAN RAINFALL INTENSITY FOR ORDINARY WET DAYS
C   HPROB    = PROBABILITY THAT A WET DAY IS HEAVY
C   HFAC     = HEAVY-EVENT RAINFALL MULTIPLIER
C   RCOEF    = DIRECT RUNOFF COEFFICIENT
C   CAP      = SOIL STORAGE CAPACITY
C   INFIL    = INFILTRATING FRACTION OF RAINFALL
C   TREND    = LINEAR INTENSITY TREND OVER ONE SYNTHETIC YEAR
C
INTEGER NDAYS, DAY, ISEED, K, WET
DOUBLE PRECISION U, RAIN, RUNOF, SOIL, PWET, WO, PERS, RMEAN
DOUBLE PRECISION HPROB, HFAC, RCOEF, ETBAS, CAP, INFIL
DOUBLE PRECISION TREND, ET, TOTR, TOTQ, MAXR, MAXQ, PI, T, SEA
DOUBLE PRECISION MEANF, EXC
PARAMETER (NDAYS=365)
DATA WO/0.200000D0/
DATA PERS/0.350000D0/
DATA RMEAN/14.000000D0/
DATA HPROB/0.080000D0/
DATA HFAC/4.000000D0/
DATA RCOEF/0.340000D0/
DATA ETBAS/3.800000D0/
DATA CAP/75.000000D0/
DATA INFIL/0.700000D0/
DATA TREND/0.120000D0/
DATA PI/3.141592653589793D0/
ISEED = 1357911
SOIL = 0.50D0*CAP
WET = 0
TOTR = 0.0D0
TOTQ = 0.0D0
MAXR = 0.0D0
MAXQ = 0.0D0
OPEN(10, FILE='results.csv', STATUS='UNKNOWN')
WRITE(10,*) 'day,rain_mm,runoff_mm,soil_mm,et_mm,wet'
DO 100 DAY = 1, NDAYS

```

```

K = ISEED / 127773
ISEED = 16807*(ISEED-K*127773)-2836*K
IF (ISEED .LE. 0) ISEED = ISEED + 2147483647
U = DBLE(ISEED)*4.656612875D-10
PWET = W0 + PERS*DBLE(WET)
IF (PWET .GT. 0.95D0) PWET = 0.95D0
RAIN = 0.0D0
IF (U .LT. PWET) THEN
  WET = 1
  K = ISEED / 127773
  ISEED = 16807*(ISEED-K*127773)-2836*K
  IF (ISEED .LE. 0) ISEED = ISEED + 2147483647
  U = DBLE(ISEED)*4.656612875D-10
  IF (U .LT. 1.0D-12) U = 1.0D-12
  T = DBLE(DAY-1)/DBLE(NDAYS)
  SEA = 1.0D0 + 0.25D0*DSIN(2.0D0*PI*(T-0.20D0))
  MEANF = RMEAN*SEA*(1.0D0+TREND*T)
  RAIN = -MEANF*DLOG(U)
  K = ISEED / 127773
  ISEED = 16807*(ISEED-K*127773)-2836*K
  IF (ISEED .LE. 0) ISEED = ISEED + 2147483647
  U = DBLE(ISEED)*4.656612875D-10
  IF (U .LT. HPROB) RAIN = RAIN*HFAC
ELSE
  WET = 0
ENDIF
T = DBLE(DAY-1)/DBLE(NDAYS)
ET = ETBAS*(1.0D0+0.35D0*DSIN(2.0D0*PI*(T-0.05D0)))
RUNOF = RCOEF*RAIN
SOIL = SOIL + INFIL*RAIN - ET
IF (SOIL .GT. CAP) THEN
  EXC = SOIL - CAP
  RUNOF = RUNOF + EXC
  SOIL = CAP
ENDIF
IF (SOIL .LT. 0.0D0) SOIL = 0.0D0
TOTR = TOTR + RAIN
TOTQ = TOTQ + RUNOF
IF (RAIN .GT. MAXR) MAXR = RAIN
IF (RUNOF .GT. MAXQ) MAXQ = RUNOF
WRITE(10,900) DAY,RAIN,RUNOF,SOIL,ET,WET
100 CONTINUE
CLOSE(10)
OPEN(11,FILE='summary.txt',STATUS='UNKNOWN')
WRITE(11,*) 'Synthetic_rainfall/runoff_demo_summary'
WRITE(11,*) 'Total_rainfall:mm:', TOTR
WRITE(11,*) 'Total_runoff:mm: ', TOTQ
WRITE(11,*) 'Maximum_rain:mm: ', MAXR
WRITE(11,*) 'Maximum_runoff:mm: ', MAXQ
CLOSE(11)
WRITE(*,*) 'Wrote_results.csv_and_summary.txt'
900 FORMAT(I4,', ',F10.3,', ',F10.3,', ',F10.3,', ',F10.3,', ',I1)
END

```